

MID SEMESTER EXAMINATION 2025

Name of the Program: B.Tech.

Semester: 1st

Name of the Course: Engineering Mathematics-I

Course Code: TMA- 101

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
 (ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Use elementary transformation to reduce the following matrix in its Echelon form and hence find the rank of the matrix $A = \begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$	CO1
OR		
(b)	Verify Cayley Hamilton theorem for matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and hence find A^{-1} .	
Q2	(10 marks)	
(a)	Using matrix method, show that the equations: $x + 2y + z = 2;$ $3x + y - 2z = 1;$ $4x - 3y - z = 3;$ $2x + 4y + 2z = 4,$ are consistent and hence obtain the solutions for x, y and z .	CO1
OR		
(b)	Examine the following vectors for linear dependence and find the relation if it exists, where $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$, and $X_4 = (-3, 7, 2)$.	
Q3	(10 marks)	
(a)	Find the Eigen values and Eigen vectors of the following matrix: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & 2 & 3 \end{bmatrix}$	CO1
OR		
(b)	Find the n^{th} derivative of $\frac{1}{x^2+a^2}$.	
Q4	(10 marks)	
(a)	If $y = 2 \cos (\log x) + 3 \sin (\log x)$, then show that $x^2 y_{n+2} + (2n+1)x y_{n+1} + (n^2+1)y_n = 0$	CO2
OR		
(b)	State Lagrange's mean value theorem and use it to prove the following: If $f(x) = \tan^{-1} x$, and $0 < a < b < 1$ prove that $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$	
Q5	(10 marks)	
(a)	Evaluate the following limit $\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{2 \sin x}$.	CO2
OR		
(b)	Using Taylor's series, obtain the value of $\cos 31^\circ$ and correct to four decimal places.	

